Code Coverage Report

for

Student Attendance Management System (SAMS)

Version: 1.0

Date: 06-11-2026

Team 1

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1. Introduction

This report gives the statement (line) coverage of the SAMS source code by the automated test suite, as asked in the testing phase. Statement coverage = lines executed by at least one test / total executable lines.

2. Tool and Command

Item	Details
Tool	coverage.py 7.17.0 (with --branch)
Command	cd sams USE_SQLITE=1 coverage run --branch --source=. manage.py test coverage report -m --omit="*/migrations/*,manage.py,sams/*,*/tests.py,*/apps.py,*/__init__.py" coverage html (same --omit)
Excluded	Migrations, Django settings / urls / wsgi (sams/), test files, apps.py, empty __init__.py
Test suite	121 tests, all passing (see Unit Test Report)
Date run	02-11-2026

3. Summary

Statements	Covered	Missed	Statement coverage
685	661	24	96.5%

Combined (statements + branches) coverage reported by coverage.py: 96.0%. Before the testing week the statement coverage was 93.1% (93 tests); 28 tests were added for untested validation paths, which raised it to 96.5%.

4. Coverage by File

File	Statements	Missed	Coverage	Missing lines
academics/admin.py	27	0	100.0%	-
academics/models.py	25	0	100.0%	-
academics/views.py	1	1	0.0%	1
accounts/admin.py	8	0	100.0%	-

accounts/decorators.py	13	0	100.0%	-
accounts/forms.py	7	0	100.0%	-
accounts/importer.py	81	0	100.0%	-
accounts/models.py	16	0	100.0%	-
accounts/urls.py	6	0	100.0%	-
accounts/views.py	23	0	100.0%	-
attendance/admin.py	7	0	100.0%	-
attendance/forms.py	19	0	100.0%	-
attendance/models.py	23	0	100.0%	-
attendance/urls.py	4	0	100.0%	-
attendance/utils.py	17	2	88.2%	28, 29
attendance/views.py	74	3	95.9%	59, 61, 62
backup_database.py	18	18	0.0%	1, 2, 3, 5, 6, 7, 8, 10, 11, 13, 14, 15,
leaves/admin.py	5	0	100.0%	-
leaves/forms.py	29	0	100.0%	-
leaves/models.py	19	0	100.0%	-
leaves/urls.py	4	0	100.0%	-
leaves/utils.py	24	0	100.0%	-
leaves/views.py	58	0	100.0%	-
reports/admin.py	1	0	100.0%	-
reports/exports.py	60	0	100.0%	-
reports/forms.py	30	0	100.0%	-
reports/models.py	1	0	100.0%	-
reports/services.py	32	0	100.0%	-
reports/urls.py	4	0	100.0%	-
reports/views.py	49	0	100.0%	-

TOTAL	685	24	96.5%	

5. Code Not Covered and Reason

File	Lines	Reason
academics/views.py	1	Empty file created by startapp (only an import), the app has no views of its own
attendance/utils.py	28, 29	Fallback when leaves.utils is missing (only needed before SEM-20 was merged)
attendance/views.py	59, 61, 62	IntegrityError branch: two faculty saving the same class at the same moment (race condition), not reproducible in a single-threaded test
backup_database.py	1, 2, 3, 5, 6, 7, 8, 10, 11, 1	Standalone backup script (NFR-4) run by hand / scheduler, not imported by the app or the tests. Checked separately in docs/evidence

6. Conclusion

96.5% of the executable application code is run by the automated tests. The remaining lines are an empty generated file, the stand-alone backup script, an import fallback and a database race condition. All requirement related code (REQ-1 to REQ-19) is covered.